



NeuroWorkflow: Agent-Assisted Brain Modeling

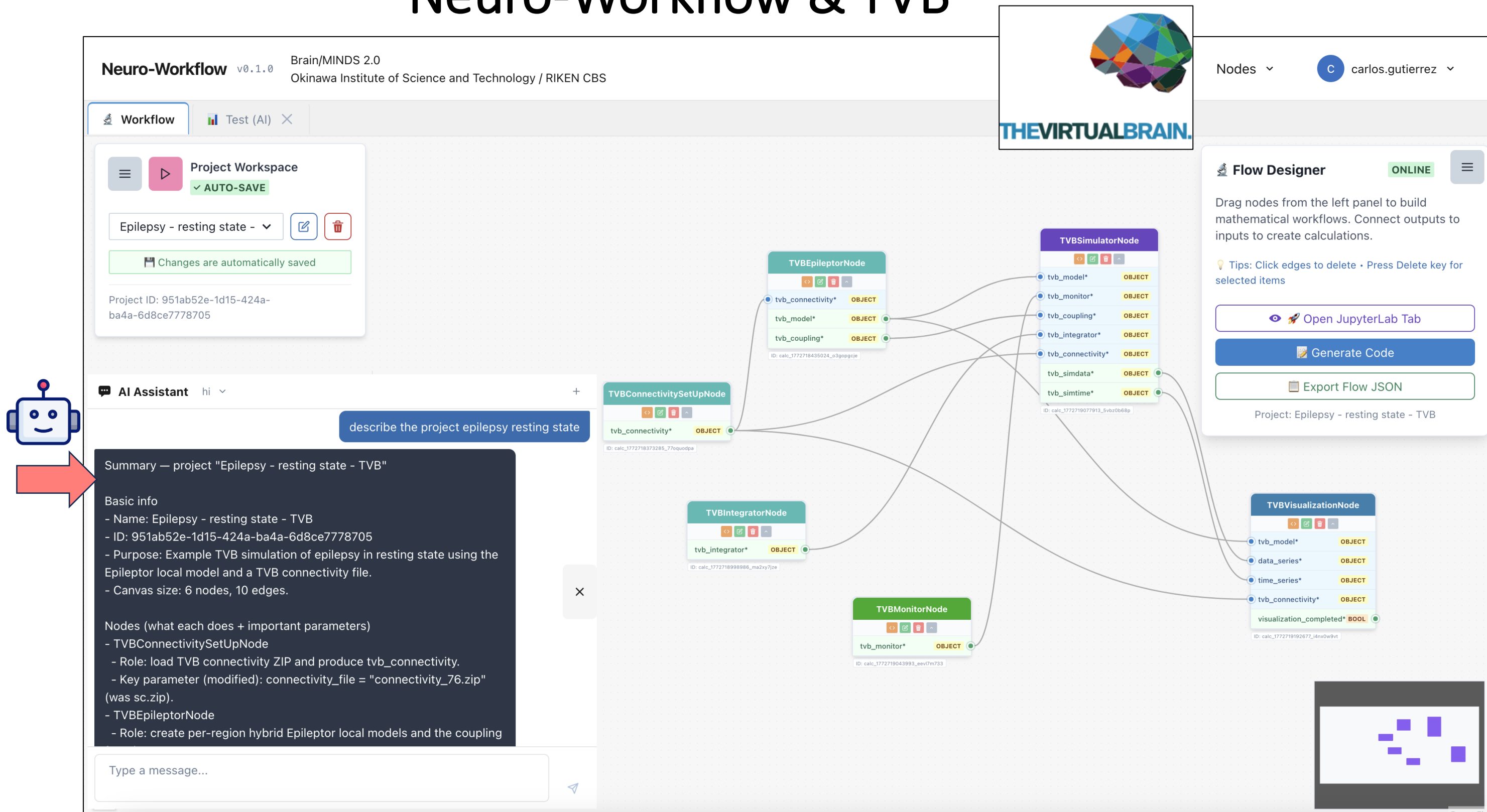


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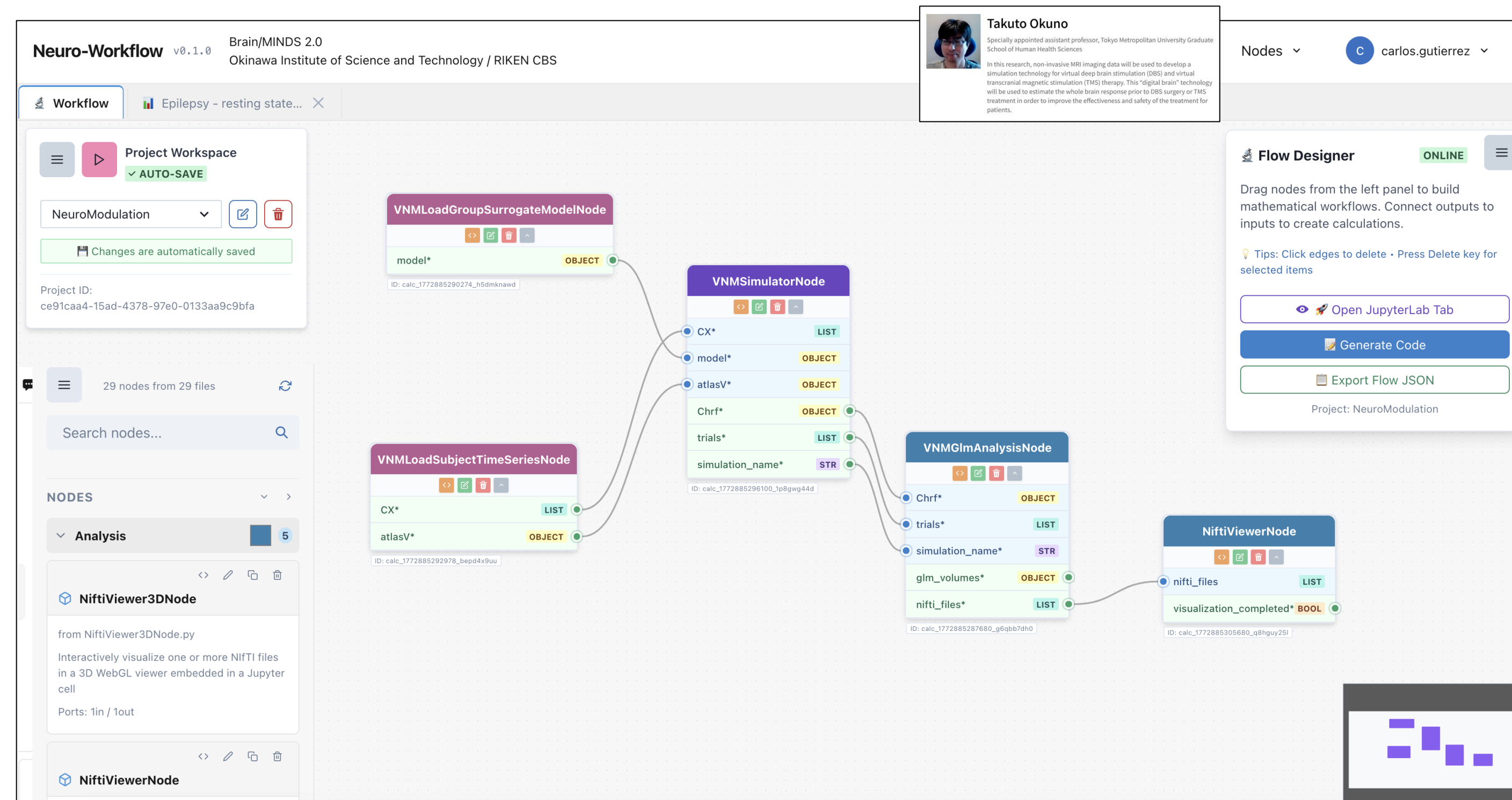
The Brain/MINDS 2.0 Neuro-Workflow

Neuro-Workflow is a node-based framework that represents brain models as self-describing graphs, making it easy for researchers, non-experts, and AI agents to assemble, reuse, and extend them. It aims to serve as the BM2 project's shared model repository. A node is a Python class wrapping a modeling step and can call modeling tools.

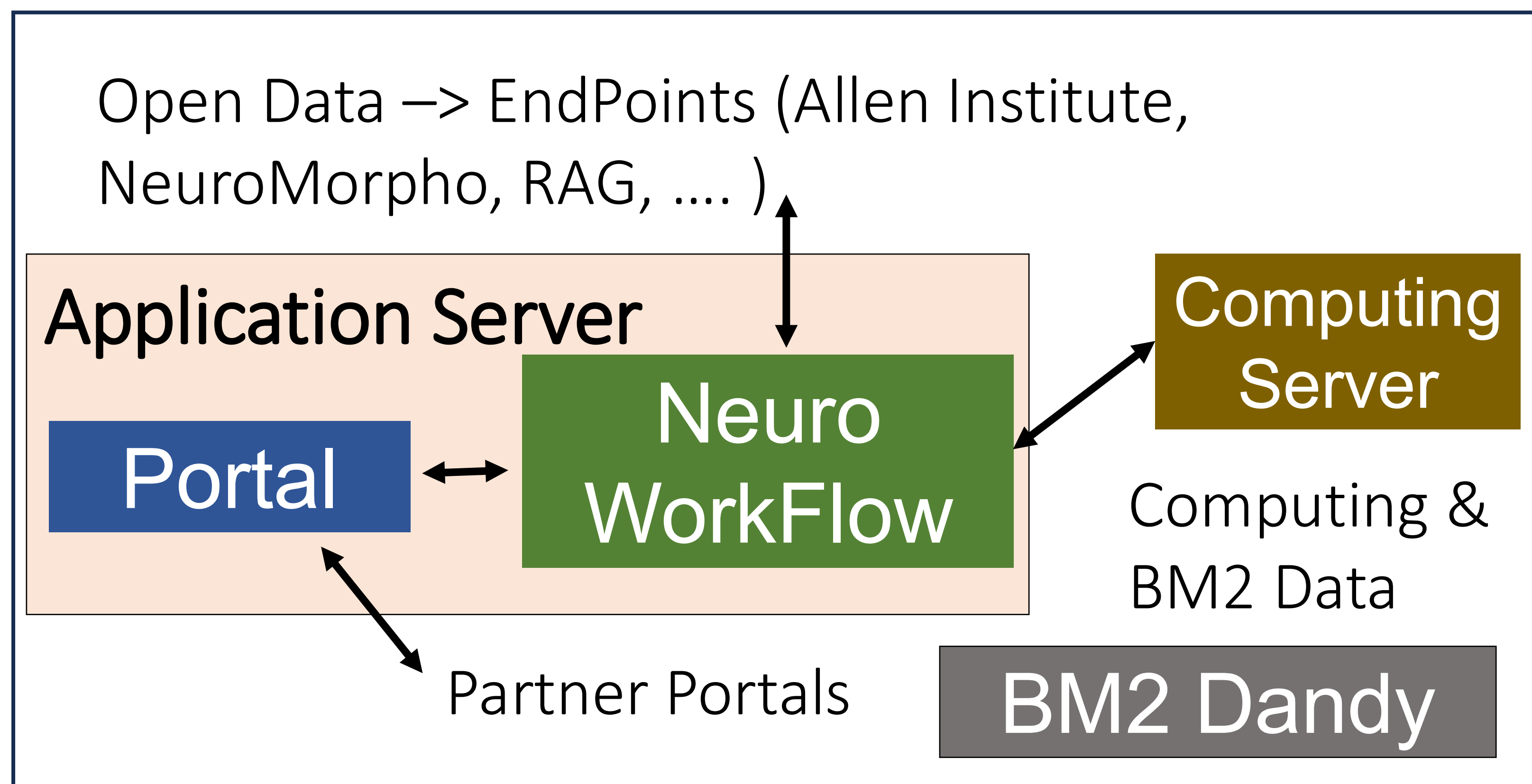
Neuro-Workflow & TVB



Neuro-Workflow & Custom code



Deployment Infrastructure



Modeling with AI

1

AI Assistant hi (Check available Nodes)

Hi — how can I help you with your NeuroWorkflow project today? (You can ask to list projects, open a workflow, add/remove nodes or edges, edit parameters, generate code, or anything else.)

I want to use the TVB for modeling, can you show me what nodes are available?

Great — here are the TVB-related nodes available (short summary of purpose, key I/O and important parameters):

- TVBConnectivitySetUpNode (category: Network)
- Purpose: load / initialize TVB structural connectivity (ZIP).
- Inputs: connectivity_file_path (optional)
- Outputs: tvb_connectivity
- Key parameter: connectivity_file (default: "sc.zip")

Type a message...

2

AI Assistant hi (Describe a Node)

provide details of the epileptor node

Here are the detailed specs for the TVBEpileptorNode:

Basic

- Label / class: TVBEpileptorNode
- File: TVBEpileptorNode.py
- Category: Network
- Description: Define the local neural (hybrid Epileptor) dynamics and coupling for TVB.

Inputs

- tvb_connectivity (object) — TVB connectivity object (optional in schema but required to initialize model realistically).

Outputs

- tvb_model (object) — configured local model (hybrid Epileptor) for each network node.

Type a message...

3

AI Assistant hi (Add a Node to working space)

connect a connectivity node, or show the raw JSON schema for copying?

add that node

Sure — which workflow/project should I add TVBEpileptorNode to? I can list your projects if you want. Also tell me where to place it (position x,y; default 0,0) and an instance name (defaults to the node label).

there is a Test (AI) project available

Done — TVBEpileptorNode was added to "Test (AI)" at position (0,0,0,0) with instance name "TVBEpileptorNode". Anything else?

LIST_PROJECTS

ADD_NODE

Type a message...

4

AI Assistant hi (Planning a workflow)

next, an epilepsy model can be made for marmoset connectivity data, what other components (nodes) should I consider?

Good — for a marmoset epilepsy model (using marmoset connectivity) you'll want nodes that cover: connectome import/conversion, model definition, integration/monitoring, simulation, visualization, and parameter tuning / analysis. Short list of recommended nodes (one line each):

- BMCR_DTI_to_Connectome — build a connectome from raw DTI/tractography (use if you start from tractography).
- BMCR_Conn_To_TVBS — convert a connectome CSV/matrix into TVB-format files / ZIP (TVB-ready).
- TVBConnectivitySetUpNode — load the TVB ZIP (or connectivity file) and produce a tvb_connectivity object.
- TVBEpileptorNode — local hybrid Epileptor model (per-region epileptogenicity heatmaps).
- TVBIntegratorNode — set integrator, dt, and noise (numerical stability /

Type a message...

5

Neuro-Workflow (Add a workflow to working space)

AI Assistant hi

I don't see the epileptor node you mentioned in your plan.

Done — I added TVBEpileptorNode to the canvas and connected TVBConnectivitySetUpNode tvb_connectivity → TVBEpileptorNode tvb_connectivity.

The workflow now contains all recommended TVB nodes and connections. Anything else you'd like to set up (parameters, add BMCR nodes, or generate code)?

Is it properly connected based on your suggestion?

Mostly — the core pipeline is in place, but two important links are missing.

What is connected (good)

- TVBConnectivitySetUpNode tvb_connectivity → TVBSimulationNode tvb_connectivity

Type a message...

6

AI Assistant hi (Generate code for simulation)

ok, good job, can you generate code for this workflow

Done — I generated Python and Jupyter notebook code for the workflow. Files saved:

- /django-app/django-project/codes/projects/Test(ai)/Test(ai).py
- /django-app/django-project/codes/projects/Test(ai)/Test(ai).ipynb

Would you like me to open or download the generated files, or run the workflow next?

GET_FLOW

GENERATE_CODE_BATCH

GENERATE_CODE_BATCH

NEW_PLAN

Type a message...

7

Python code

```
#!/usr/bin/env python3
import sys
import os

# Add paths for JupyterLab environment
sys.path.append('...')

from neuroworkflow.core.workflow import WorkflowBuilder
from nodes.network.TVBEpileptorNode import TVBEpileptorNode
from nodes.analysis.TVBVisualizationNode import TVBVisualizationNode
from nodes.network.TVBConnectivitySetUpNode import TVBConnectivitySetUpNode
from nodes.network.TVBIntegratorNode import TVBIntegratorNode
from nodes.simulation.TVBSimulationNode import TVBSimulationNode
from nodes.stimulus.TVBStimulusNode import TVBStimulusNode

def main():
    """Run a simple neural simulation workflow."""
    # workflow_builder_import
    # workflow_builder_creation
    workflow_builder = WorkflowBuilder(
        "Test (AI)",
        context={}
    )

    # Create nodes
    instance_TVBConnectivitySetUpNode_001 = TVBConnectivitySetUpNode("instance_TVBConnectivitySetUpNode_001",
                                                                     connectivity_file_path="sc.zip")
    instance_TVBIntegratorNode_002 = TVBIntegratorNode("instance_TVBIntegratorNode_002",
                                                       dt=0.01, noise=0.001)
    instance_TVBSimulationNode_003 = TVBSimulationNode("instance_TVBSimulationNode_003",
                                                         tvb_connectivity=instance_TVBConnectivitySetUpNode_001,
                                                         tvb_integrator=instance_TVBIntegratorNode_002,
                                                         tvb_stimulus=instance_TVBStimulusNode_004)
    instance_TVBSimulationNode_003.run()

    # Visualization
    instance_TVBVisualizationNode_004 = TVBVisualizationNode("instance_TVBSimulationNode_003",
                                                             tvb_simulation=instance_TVBSimulationNode_003)
    instance_TVBVisualizationNode_004.run()

    # Save results
    instance_TVBVisualizationNode_004.save()

    # Clean up
    instance_TVBConnectivitySetUpNode_001.cleanup()
    instance_TVBIntegratorNode_002.cleanup()
    instance_TVBSimulationNode_003.cleanup()
    instance_TVBVisualizationNode_004.cleanup()

if __name__ == '__main__':
    main()
```